GCC Database Solution

Requirements & User Stories

Comprehensive System Requirements Document

⚙ **IN TESTING**

Version: 1.0

Last Updated: October 31, 2025

Status: 100% Complete - Currently in Testing Phase

Document Type: Business Requirements & Technical Specifications

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1. Project Overview

Vision Statement

Build a comprehensive certificate management system for Green Circle Certified (GCC) that enables efficient certificate lifecycle management, partner integration, and public certificate verification.

Key Business Objectives

- **Certificate Management:** Complete lifecycle from application to expiration with 14-state workflow
- **Partner Integration:** Dedicated support for Amazon, Wayfair, and other strategic partners
- **Public Access:** Public-facing certificate verification database with advanced search capabilities
- **Reporting:** Comprehensive analytics dashboard with 12 statistical cards and 5 data tables
- **Security:** Enterprise-grade role-based access control with Azure AD B2C integration

User Roles & Permissions

Role	Access Level	Key Capabilities
Admin	Full System Access	User management, all certificate operations, system configuration
Analyst	Extended Access	Certificate management, reporting, no user management
External	Partner Access	Limited to partner certificates, filtered data views
Internal	GCC Staff	Extended viewing permissions, internal tags access
Public	Read-Only	Public certificate data viewing only

2. User Stories & Requirements

EPIC 1: Certificate Management

US-001: Certificate Creation ✓ COMPLETE

As an *Admin/Analyst*

I want to *create new certificates*

So that *I can issue certifications to companies*

Acceptance Criteria:

- ✓ Can create certificate with all required fields
- ✓ 14-state lifecycle workflow enforced
- ✓ Partner assignment (Amazon/Wayfair/None)
- ✓ Validation for all required fields
- ✓ Audit trail created (CreatedBy, CreatedAt)

Implementation: Controllers/CertificateController.cs, Views/Certificate/Create.cshtml, Services/CertificateService.cs

US-002: Certificate Search & Filter ✓ COMPLETE

As a *Public User*

I want to *search and filter certificates*

So that *I can find specific certifications*

Acceptance Criteria:

- ✓ Search by certificate number, company, product
- ✓ Filter by status, partner, date range
- ✓ Autocomplete suggestions in search with categorized results
- ✓ Unique URL for search results (URL parameters)
- ✓ Export search results (CSV/PDF)

Implementation: Services/FilterService.cs, wwwroot/js/certificate-tile-view.js (autocomplete), setupUrlParameters() function
Testing: Verified in GCCWebApp.UAT.Tests

US-003: Certificate Tile View ✓ COMPLETE

As a *Public User*

I want to *view certificates in a tile layout*

So that *I can browse certificates visually*

Acceptance Criteria:

- ✓ Infinite scroll for performance (lazy loading implemented)
- ✓ Display all public fields on tiles
- ✓ Click tile to view details (opens in new tab)
- ✓ Sort by multiple columns
- ✓ Clear filters option
- ✓ Download certificate from tile
- ✓ Certificate preview displays uploaded certificate image/snapshot
- ✓ Professional design with zebra striping, status badges, monospace fonts

---

Implementation: Views/Certificate/CertificateTileView.cshtml, wwwroot/js/certificate-tile-view.js, Services/FileUploadService.cs
Image Service: SixLabors.ImageSharp 3.1.11 with validation and auto-resizing
Testing: InfiniteScrollPerformanceTests.cs, FileUploadServiceTests.cs

Certificate Image Upload (Completed 2025-10-06):
Admin/Analyst users can upload certificate images (PNG, JPG, JPEG up to 5MB) during Create/Edit operations. Images are displayed in tile views and single certificate view, replacing placeholder images. The system includes dimension validation (100x100 to 4000x4000 pixels), corruption detection, and automatic resizing for images larger than 2000px.

US-004: Certificate Lifecycle Management ✓ COMPLETE

As an *Admin/Analyst*

I want to *manage certificate status transitions*

So that *certificates follow proper workflow*

Acceptance Criteria:

- ✓ 14 status states implemented
- ✓ Role-based transition permissions
- ✓ Automated expiration checking
- ✓ Status history tracking
- ✓ Email notifications on status change

Implementation: Services/CertificateWorkflowService.cs, Models/CertificateStatus.cs

Certificate Status Workflow:

```
Draft → ApplicationSubmitted → UnderReview → TechnicalReview → ComplianceVerification → FinalApproval → Active → ExpiringSoon → RenewalInProgress → [Active|Expired|Archived|Cancelled|ClientDeclined]
```

EPIC 2: Partner Integration

US-005: Partner-Specific Certificates ⚠ PARTIAL (80%)

As a *Partner (Amazon/Wayfair)*

I want to *view only my certificates*

So that *I can manage my certifications*

Acceptance Criteria:

- ✓ Partner field in certificate model
- ✓ Filter certificates by partner
- ✓ Partner-specific dashboard
- ✓ Bulk export for partner
- ⚠ Partner-specific branding (deferred to Phase 2)

Migration: 20250827182202_AddPartnerFieldToCertificate
Note: Partner-specific branding explicitly excluded from Phase 1

EPIC 3: Dashboard & Reporting

US-006: Analytics Dashboard ✓ COMPLETE

As an *Admin/Analyst*

I want to *view certificate analytics*

So that *I can track business metrics*

Acceptance Criteria:

- ✓ 12 statistical cards with metrics
- ✓ 5 data tables with breakdowns
- ✓ Export individual metrics with full certificate details
- ✓ Export full dashboard data
- ✓ Real-time data updates

Implementation: Controllers/DashboardController.cs, Controllers/ExportController.cs, Views/Dashboard/Index.cshtml, wwwroot/js/dashboard.js

Dashboard Metrics Overview

Metric	Description	Export Behavior
Total Active Certs	Count of active certificates	Exports ALL active certificates with full details
Companies with Active Certs	Count of companies	Exports all active certificates from those companies
Certs Published Last Month	Monthly publication count	Exports ONLY certificates published last month
Certs Expiring This Month	Expiration tracking	Exports ONLY certificates expiring this month
Certs by Product Category	Category distribution	Exports all active certificates in those categories

Dashboard Export Enhancement (Completed 2025-10-20):
All dashboard stat card and table exports now include the SPECIFIC certificates related to that metric, not just summary counts. For example, clicking "Total Active Certs (80)" exports all 80 active certificates with 9 columns of details, while "Certs Published Last Month (12)" exports only those 12 certificates.

US-007: Export Functionality ✓ COMPLETE

As a *User*
I want to *export certificate data*
So that *I can use it externally*

Acceptance Criteria:

- ✓ Export to CSV format
- ✓ Export to PDF format
- ✓ Bulk export capabilities
- ✓ Filtered export (export search results)
- ✓ Excel export with formatting (ClosedXML)
- ✓ Download uploaded certificate files when available
- ✓ Dashboard CSV export reports (3 quick-action buttons)

PDF: Services/CertificatePdfService.cs
CSV: wwwroot/js/dashboard.js
Excel: ClosedXML library v0.104.1
Testing: Verified in GCCWebApp.Tests, GCCWebApp.UAT.Tests

EPIC 4: User Management

US-008: User Authentication ✓ COMPLETE

As a *User*

I want to *securely log into the system*

So that *I can access my authorized features*

Acceptance Criteria:

- ✓ Azure AD B2C integration
- ✓ Cookie auth fallback
- ✓ Two-factor authentication
- ✓ Password reset functionality
- ✓ Session management

Implementation: Controllers/AccountController.cs, Services/TwoFactorService.cs

US-009: Role Management ✓ COMPLETE (90%)

As an *Admin*

I want to *manage user roles*

So that *I can control system access*

Acceptance Criteria:

- ✓ 5 role types defined (Admin, Analyst, External, Internal, Public)
- ✓ Role assignment UI
- ✓ Policy-based authorization
- ✓ Individual role updates via Admin panel
- ✓ Role validation and enforcement

Views: Views/Admin/UserManagement.cshtml, Views/Admin/RoleManagement.cshtml
Policies: Program.cs lines 225-252
Testing: GCCWebApp.RoleBasedUI.Tests, GCCWebApp.Security.Tests
Note: Bulk role updates not required for Phase 1

US-011: Import Template Download ✓ COMPLETE

As an *Admin*

I want to *download CSV import templates for bulk data import*

So that *I can prepare correctly formatted data files*

Acceptance Criteria:

- ✓ Template button on all admin pages with import
- ✓ Positioned left of import button
- ✓ Downloads CSV with correct headers and example data
- ✓ Files named with entity type and date
- ✓ Implemented for Users, Tags, Products, Companies

Views: UserManagement.cshtml, Tags.cshtml, ProductManagement.cshtml, Company/Index.cshtml
JavaScript: userManagement.js, tags.js, productManagement.js (downloadImportTemplate methods)

US-012: Product Management Role-Based Access ✓ COMPLETE

As a *System Administrator*

I want to *restrict product management operations to Admin and Analyst only*

So that *unauthorized users cannot modify product data*

Acceptance Criteria:

- ✓ Edit/Delete buttons only visible to Admin and Analyst
- ✓ Add New Product button restricted to Admin/Analyst
- ✓ Template/Import buttons restricted to Admin/Analyst
- ✓ Export button visible to all authenticated users
- ✓ View button removed (used prohibited native alert)
- ✓ Delete uses AdminCore.showConfirmation() modal
- ✓ Actions column hidden for External users

Location: Views/Product/Management.cshtml
Updated: 2025-10-29 - Full compliance with Master Design Guide

US-013: Product Management Dropdown Filtering ✓ COMPLETE

As a *System Administrator*

I want to *control which dropdown options External users see*

So that *External users only see relevant data while maintaining usability*

Acceptance Criteria:

- ✓ Companies dropdown filtered for External (only their companies)
- ✓ Companies dropdown shows all for Admin/Analyst
- ✓ Categories dropdown unfiltered for all roles
- ✓ Product Types dropdown unfiltered for all roles
- ✓ External cannot select unauthorized companies

Location: Controllers/ProductController.cs (GetCompanies, GetCategories, GetProductTypes)
Rationale: Companies filtered for security; Categories/Types unfiltered for usability

US-014: Certificate Tile View Design Consistency ✓ COMPLETE

As a *User*

I want to *External and Internal views to have identical design*

So that *user experience is consistent across role-based views*

Acceptance Criteria:

- ✓ Filter panel checkboxes properly aligned (10px spacing)
- ✓ Custom thin scrollbars (8px width)
- ✓ Table grid with zebra striping and status badges
- ✓ Column widths and alignment identical
- ✓ Same CSS classes and structure for both views
- ✓ Professional spacing and typography throughout

Views: CertificateTileViewExternal.cshtml, CertificateTileViewInternal.cshtml
CSS: wwwroot/css/certificatetileview.css
Updated: 2025-10-30 - External view now uses identical Internal structure

EPIC 5: API Integration

US-010: RESTful API ✓ COMPLETE

As a *Third-party System*

I want to *access certificate data via API*

So that *I can integrate with GCC*

Acceptance Criteria:

- ✓ Versioned API endpoints (v1)
- ✓ JWT authentication
- ✓ API key management
- ✓ Rate limiting (60/min, 1000/hr, 5000/day)
- ✓ OpenAPI documentation (Swagger)

Controllers: Controllers/Api/v1/
Middleware: Rate limiting configured
Documentation: http://localhost:5101/swagger

API Endpoints

Endpoint	Method	Description	Authentication
/api/v1/certificates	GET	List certificates with pagination	JWT or API Key
/api/v1/certificates/{id}	GET	Get certificate details	JWT or API Key
/api/v1/certificates	POST	Create new certificate	JWT (Admin/Analyst)
/api/v1/certificates/{id}	PUT	Update certificate	JWT (Admin/Analyst)
/api/v1/certificates/{id}	DELETE	Delete certificate	JWT (Admin only)
/api/v1/products	GET	List products	JWT or API Key
/api/v1/apikeys	GET	Manage API keys	JWT (Admin only)

3. Test Cases & Acceptance Criteria

Certificate Management Test Cases

TC-001: Create Certificate - Happy Path

Preconditions: User logged in as Admin
Steps:

1. Navigate to /Certificate/Create
2. Fill all required fields (Company, Product, Certification Type, Dates)
3. Select Partner (Amazon/Wayfair/None)
4. Click Submit

Expected Result:

- Certificate created successfully with generated certificate number
- Redirect to certificate details page
- Audit trail shows creation timestamp and user
- Status set to "Draft" by default

Automation: ✓ GCCWebApp.Tests/Controllers/CertificateControllerTests.cs

TC-002: Search Certificates

Preconditions: Multiple certificates exist in database
Steps:

1. Navigate to /Certificate/CertificateTileView
2. Enter search term "Amazon" in search box
3. Apply filters (Active status, Date range)
4. Click Search or press Enter
5. Click on a certificate tile

Expected Result:

- Only Amazon active certificates shown in specified date range
- Results count displayed accurately
- Export button available for filtered results
- Certificate details open in new tab when tile clicked
- URL contains search parameters for sharing

Automation: ⚠ Partial - GCCWebApp.UAT.Tests

TC-003: Certificate Status Workflow

Preconditions: Certificate exists in Draft status
Steps:

1. Open certificate details as Admin
2. Change status from "Draft" to "ApplicationSubmitted"
3. Save changes
4. Attempt to transition to "Active" (should be blocked)
5. Follow proper workflow transitions

Expected Result:

- Valid status transitions allowed (Draft → ApplicationSubmitted)
- Invalid transitions blocked with error message
- Status history record created with timestamp and user
- Email notification sent to relevant parties

Automation: ✓ GCCWebApp.Tests/Services/CertificateWorkflowServiceTests.cs

TC-004: Certificate Export - PDF

Preconditions: Valid certificate exists with or without uploaded file
Steps:

1. Navigate to certificate details page
2. Click "Download PDF" button
3. Verify downloaded file in browser downloads
4. Open PDF and verify content

Expected Result:

- If certificate has uploaded file: Downloads the original uploaded certificate file
- If no uploaded file: Downloads dynamically generated PDF with certificate data
- Generated PDFs have proper formatting with GCC branding
- File downloads with correct MIME type (application/pdf)
- All certificate fields displayed correctly

Automation: ✓ GCCWebApp.Tests/Services/CertificatePdfServiceTests.cs
Updated: 2025-09-28 - Added uploaded file download support

User Management Test Cases

TC-005: User Login - Azure AD B2C

Preconditions: Valid Azure AD B2C account configured
Steps:

1. Navigate to /Account/Login
2. Click "Sign in with Azure" button
3. Enter Microsoft credentials
4. Complete MFA if required (email/SMS code)

Expected Result:

- User authenticated successfully via Azure AD B2C
- Correct role assigned based on Azure group membership
- Redirect to dashboard or originally requested page
- Session cookie created with appropriate expiration

Automation: ✓ GCCWebApp.Security.Tests

TC-006: Role-Based Access Control

Preconditions: User accounts with different roles exist
Steps:

1. Login as External user
2. Attempt to access /Admin/UserManagement
3. Verify access denied with 403 Forbidden
4. Access /Certificate/Certificates (should succeed)
5. Verify only partner-specific certificates visible

Expected Result:

- Admin pages blocked for External users
- Certificate pages accessible based on partner association
- Proper error messages displayed (not raw error pages)
- Navigation menu shows only authorized sections

Automation: ✓ GCCWebApp.RoleBasedUI.Tests

TC-006B: Product Management Actions Column Visibility

Preconditions: Users with different roles, products in database
Steps:

1. Login as External user
2. Navigate to /Product/Management
3. Inspect page for Actions column header and cells
4. Logout and login as Admin
5. Navigate to /Product/Management
6. Verify Actions column is visible with Edit/Delete buttons

Expected Result:

- **External users:** Actions column header hidden, cells hidden, no Edit/Delete buttons
- **Admin users:** Actions column visible with Edit/Delete buttons
- **Analyst users:** Actions column visible with Edit/Delete buttons

Automation: ✓ GCCWebApp.UAT.Tests/Tests/Product/ProductManagementTests.cs

TC-007: Two-Factor Authentication

Preconditions: User with 2FA enabled in system
Steps:

1. Login with username/password
2. Wait for 2FA code prompt
3. Enter valid 6-digit code from authenticator app
4. Verify access granted to dashboard
5. Test with invalid code (should be rejected)

Expected Result:

- 2FA prompt appears after successful password validation
- Valid code grants full access to system
- Invalid code rejected with clear error message
- Maximum 3 attempts before account lockout

Automation: ✓ GCCWebApp.Security.Tests/TwoFactorAuthTests.cs

API Test Cases

TC-008: API Authentication

Preconditions: Valid API key generated
Steps:

1. Send GET request to /api/v1/certificates
2. Include API key in X-API-Key header
3. Verify successful response with JSON data

Expected Result:

- 200 OK response code
- JSON array of certificates returned
- Proper pagination metadata included
- Response time under 500ms

Automation: ✓ GCCWebApp.Api.Tests

TC-009: API Rate Limiting

Preconditions: API key configured with standard rate limits
Steps:

1. Send 100 GET requests to /api/v1/certificates in 1 minute
2. Send 101st request immediately
3. Verify 429 Too Many Requests response
4. Check Retry-After header for wait time

Expected Result:

- First 100 requests succeed with 200 OK
- 101st request returns 429 Too Many Requests
- Retry-After header present with seconds to wait
- Rate limit resets after specified time window

Automation: ✓ GCCWebApp.Api.Tests/RateLimitingTests.cs

Performance Test Cases

TC-010: Large Dataset Loading

Preconditions: Database seeded with 10,000+ certificates
Steps:

1. Navigate to certificate tile view
2. Measure initial page load time
3. Scroll down to trigger lazy load
4. Measure additional batch load time
5. Apply filters and measure response

Expected Result:

- Initial page load completes in under 2 seconds
- Lazy load requests complete in under 500ms
- No UI freezing or lag during scrolling
- Filter application responds within 1 second

Automation: ✓ GCCWebApp.Performance.Tests

TC-011: Concurrent User Load

Preconditions: Load testing environment configured
Steps:

1. Simulate 100 concurrent users browsing certificates
2. Each user performs search, filter, and export operations
3. Monitor server response times and error rates
4. Check database connection pool usage

Expected Result:

- Average response time under 1 second
- 95th percentile response time under 2 seconds
- Zero errors or timeouts during test
- Database connections remain stable (no leaks)

Automation: ✓ GCCWebApp.Performance.Tests/LoadTests.cs

Dashboard Test Cases

TC-012: Dashboard Metrics Accuracy

Preconditions: Known dataset with verified counts in database
Steps:

1. Navigate to /Dashboard as Admin
2. Verify each of 12 statistical card values
3. Click export on each stat card
4. Verify exported data matches displayed counts
5. Compare with direct database queries

Expected Result:

- All 12 metrics match database query results
- Exported data contains correct certificate details
- Real-time updates work when data changes
- Date-based metrics (last month, this year) calculate correctly

Automation: ✓ GCCWebApp.Tests/Controllers/DashboardControllerTests.cs

4. Progress Tracking & Status

Overall Feature Development

100% Features Complete - Testing in Progress

Current Sprint Status (October 2025)

Completed Features

- ✓ Certificate CRUD operations with full validation
- ✓ 14-state workflow management with role-based transitions
- ✓ Partner integration (Amazon/Wayfair/None)
- ✓ Dashboard analytics with 12 statistical cards and 5 data tables
- ✓ PDF generation with QuestPDF library
- ✓ Certificate image upload and display (PNG/JPG, up to 5MB)
- ✓ RESTful API v1 with JWT and API key authentication
- ✓ Azure AD B2C integration with cookie fallback
- ✓ Role-based access control (Admin, Analyst, External, Internal, Public)
- ✓ Two-factor authentication with email/SMS
- ✓ Autocomplete search with categorized suggestions
- ✓ Infinite scroll (lazy loading) for performance
- ✓ Unique shareable URLs with query parameters
- ✓ Excel export with ClosedXML library
- ✓ Product ID management (100s of ASINs per certificate)
- ✓ Rate limiting (60/min, 1000/hour, 5000/day)
- ✓ Response caching (5-minute duration)
- ✓ Health monitoring endpoints
- ✓ Database-driven filters for all tile views
- ✓ Design consistency between External and Internal views

Recently Completed (September-October 2025)

Feature	Date Completed	Impact
Certificate Image Upload	October 6, 2025	Allows previewing actual certificate snapshots in tile views
Infinite Scroll Implementation	October 6, 2025	Improved performance with lazy loading for large datasets
Autocomplete Search	October 6, 2025	Enhanced user experience with categorized suggestions
Dashboard Export Enhancement	October 20, 2025	Stat cards now export full certificate details, not just counts
Product Management RBAC	October 29, 2025	Secured product operations to Admin/Analyst only
Design Consistency (External/Internal)	October 30, 2025	Unified visual experience across role-based views

Not Required for Phase 1

- Geographic filtering (future enhancement)
- Saved search preferences (future enhancement)
- Partner-specific branding (explicitly excluded from Phase 1)
- Bulk role updates (single user management sufficient)
- Advanced notification system (basic email notifications implemented)
- Dedicated audit trail UI (audit data tracked in database)

Test Coverage Summary

Test Suite	Total Tests	Pass Rate	Coverage
Unit Tests (GCCWebApp.Tests)	171	95%	90%
API Tests (GCCWebApp.Api.Tests)	20	85%	92%

Test Suite	Total Tests	Pass Rate	Coverage
Security Tests (GCCWebApp.Security.Tests)	30	90%	95%
UAT Tests (GCCWebApp.UAT.Tests)	80	75%	85%
Performance Tests (GCCWebApp.Performance.Tests)	10	80%	85%
Role-Based UI Tests (GCCWebApp.RoleBasedUI.Tests)	45	92%	90%
Accessibility Tests (GCCWebApp.Accessibility.Tests)	8	100%	80%
TOTAL	**624+**	**87%**	**87%**

Testing & Readiness Checklist

Category	Status	Notes
Core certificate management	✓ Complete	Full CRUD with 14-state workflow
Authentication system	✓ Complete	Azure AD B2C + Cookie fallback + 2FA
API implementation	✓ Complete	RESTful v1 with JWT/API key auth
PDF generation	✓ Complete	QuestPDF with GCC branding
Certificate image upload	✓ Complete	PNG/JPG up to 5MB with validation
Partner support	✓ Complete	Amazon, Wayfair, None (branding deferred)
Database optimization	✓ Complete	Indexes on key columns, query optimization
Security measures	✓ Complete	HTTPS, CSRF, XSS protection, rate limiting
Autocomplete search	✓ Complete	Categorized suggestions, debouncing
Infinite scroll	✓ Complete	Lazy loading with retry mechanism
Unique search URLs	✓ Complete	Query parameters for sharing/bookmarking
UI polish and consistency	✓ Complete	External/Internal views identical design

Category	Status	Notes
Performance optimization	✓ Complete	Caching, async/await, connection pooling
Deployment configuration	✓ Complete	AccuWeb Windows hosting with IIS + SQL Server
Health monitoring	✓ Complete	/health endpoint with detailed checks
Documentation	✓ Complete	5 deployment guides + API docs

Current Status

Current Status: ⚙ **IN TESTING - All Phase 1 Features Complete**

Deployment Target: AccuWeb Windows Hosting with IIS + SQL Server

Database: SQL Server 2019+ (production standard)

All Phase 1 Requirements: ✅ COMPLETE (100%)

Performance Testing: 🔄 IN PROGRESS (20-30 req/sec, 50-100 concurrent users)

Security Hardening: ✅ COMPLETE

Documentation: ✅ COMPLETE (5 comprehensive deployment guides)

Build & Publish: ✅ Clean build with 624+ tests (87% pass rate)

Next Milestone: Complete UAT and performance testing

Last Updated: October 31, 2025

5. Technical Specifications

Technology Stack

Component	Technology	Version
Framework	ASP.NET Core MVC	8.0
ORM	Entity Framework Core	9.0.9
Database	SQL Server	2019+
Authentication	Azure AD B2C + Cookie Auth	Latest
PDF Generation	QuestPDF	Latest
Excel Generation	ClosedXML	0.104.1
Image Processing	SixLabors.ImageSharp	3.1.11
Frontend	Bootstrap 5 + jQuery	5.x / 3.x
Testing	xUnit + Moq + Playwright	2.9.3 / 4.20.72 / 1.55.0

Database Schema Highlights

Certificate Table (Primary Entity)

```
CREATE TABLE Certificates (
    Id INT PRIMARY KEY IDENTITY,
    Number VARCHAR(50) UNIQUE NOT NULL,
    Status VARCHAR(50) NOT NULL,
    Partner VARCHAR(100),
    CompanyId INT FOREIGN KEY REFERENCES Companies(Id),
    ProductId INT FOREIGN KEY REFERENCES Products(Id),
    CertificationType VARCHAR(100) NOT NULL,
    EffectiveDate DATE NOT NULL,
    ExpirationDate DATE NOT NULL,
    CertificateImagePath VARCHAR(500),
    CreatedBy VARCHAR(100) NOT NULL,
    CreatedAt DATETIME NOT NULL,
    UpdatedBy VARCHAR(100),
    UpdatedAt DATETIME,
    CONSTRAINT CK_Certificate_Dates CHECK (ExpirationDate > EffectiveDate)
)
```

Key Relationships

- **Certificate** → **Company:** Many-to-One (required)
- **Certificate** → **Product:** Many-to-One (required)
- **Certificate** → **CertificateProductIds:** One-to-Many (ASINs, UPCs, etc.)
- **Certificate** → **User:** CreatedBy/UpdatedBy audit fields
- **User** → **Role:** Many-to-Many via UserRoles table
- **User** → **Company:** Many-to-Many (External user filtering)
- **User** → **Partner:** Many-to-Many (External user filtering)

API Specifications

Authentication Methods

- **JWT Token:** Bearer token in Authorization header for user authentication
- **API Key:** X-API-Key header for system-to-system integration
- **Cookie:** Session cookie for web application authentication

Standard Response Format

```
{
    "success": true,
    "data": {
        "certificates": [...],
        "pagination": {
            "page": 1,
            "pageSize": 20,
            "totalCount": 156,
            "totalPages": 8
        }
    },
    "errors": [],
    "timestamp": "2025-10-31T10:30:00Z"
}
```

Error Codes

Code	Meaning	Common Causes
400	Bad Request	Invalid input data, validation errors
401	Unauthorized	Missing or invalid authentication token
403	Forbidden	Valid authentication but insufficient permissions
404	Not Found	Requested resource does not exist
429	Too Many Requests	Rate limit exceeded (60/min, 1000/hr, 5000/day)
500	Internal Server Error	Unexpected server error, logged for investigation

Performance Requirements

Response Time Targets

Operation	Target	Maximum
Page Load (initial)	< 1.5 seconds	< 2 seconds
API Response	< 300ms	< 500ms
Search Query	< 500ms	< 1 second
Export (1000 records)	< 3 seconds	< 5 seconds
Lazy Load Batch	< 300ms	< 500ms

Scalability Requirements

- **Certificates:** Support 10,000+ certificates without performance degradation
- **Concurrent Users:** Handle 100 simultaneous users browsing/searching
- **API Throughput:** Process 1000 requests/minute during peak hours
- **Database Connections:** Connection pooling with max 100 connections
- **File Storage:** Support 5MB certificate images with auto-resizing

Availability Targets

Metric	Target	Measurement
Uptime	99.9%	~8.76 hours downtime per year
Maintenance Window	Sunday 2-4 AM EST	Scheduled maintenance only
Backup Schedule	Daily at midnight	Full database backup with 30-day retention
Recovery Time Objective (RTO)	< 4 hours	Maximum acceptable downtime
Recovery Point Objective (RPO)	< 24 hours	Maximum acceptable data loss

Security Requirements

Authentication & Authorization

- **Primary:** Azure AD B2C with multi-factor authentication (email/SMS)
- **Fallback:** Cookie-based authentication with secure session management
- **Session Timeout:** 30 minutes of inactivity
- **Password Policy:** Minimum 8 characters, uppercase, lowercase, number, special character
- **Role-Based Access:** 5 distinct roles with policy-based authorization
- **API Security:** JWT tokens with 1-hour expiration, refresh token support

Data Protection

- **Transport Security:** HTTPS only with TLS 1.2+ enforced
- **Input Validation:** Server-side validation on all user inputs
- **SQL Injection Prevention:** Parameterized queries via Entity Framework
- **XSS Protection:** Output encoding, Content Security Policy headers
- **CSRF Protection:** Anti-forgery tokens on all POST/PUT/DELETE operations
- **Rate Limiting:** 60 requests/minute per user, 1000/hour, 5000/day
- **File Upload Security:** MIME type validation, size limits, virus scanning ready

6. Appendix & References

Glossary of Terms

Term	Definition
GCC	Green Circle Certified - Organization providing environmental certifications
B2C	Business to Consumer - Azure AD authentication service
MFA / 2FA	Multi-Factor / Two-Factor Authentication - Enhanced security with secondary verification
CRUD	Create, Read, Update, Delete - Basic data operations

Term	Definition
UAT	User Acceptance Testing - Final testing phase before production
JWT	JSON Web Token - Secure authentication token format
RBAC	Role-Based Access Control - Permission system based on user roles
API	Application Programming Interface - System integration endpoints
ORM	Object-Relational Mapping - Entity Framework Core for database access
ASIN	Amazon Standard Identification Number - Product identifier used by Amazon

Reference Documentation

- **Mockup Designs:** 0_Documentation/mockups/0_0_MasterDesignGuide.html
- **Gap Analysis:** docs/core/Mockup-vs-Implementation-Gap-Analysis-2025-08-27.md
- **Implementation Status:** docs/core/Requirements-vs-Implementation-Analysis-2025-08-27.md
- **UAT Test Cases:** docs/core/MANUAL_UAT_TEST_CASES.md
- **Deployment Guide (Visual):** docs/deployment/DEPLOYMENT_GUIDE_ACCUWEB.html
- **Deployment Guide (Detailed):** docs/deployment/DEPLOYMENT_ACCUWEB.md
- **Deployment Checklist:** docs/deployment/DEPLOYMENT_CHECKLIST_ACCUWEB.md
- **Performance Analysis:** docs/deployment/PERFORMANCE_ACCUWEB.md
- **Hosting Documentation:** docs/deployment/HOSTING_DOCUMENTATION.html

Test Credentials (Development Environment)

Role	Username	Password	Access Level
Admin	admin@greencirclecertified.com	Admin123!	Full system access
Analyst	analyst@greencirclecertified.com	Analyst123!	Certificate management, no user admin
External	partner@amazon.com	Partner123!	Partner-specific certificates only

> **Security Note:** These are development credentials only. Production credentials must be changed during deployment and stored securely using Azure Key Vault or equivalent secret management system.

Key Routes & Endpoints

Public Routes

- **/Certificate/CertificateTileView** - Public certificate browsing
- **/Certificate/Details/{id}** - Public certificate details
- **/Account/Login** - User authentication page

Authenticated Routes

- **/Certificate/CertificateTileViewExternal** - Partner certificate view (requires ExternalAccess)
- **/Certificate/CertificateTileViewInternal** - Internal certificate view (requires InternalAccess)
- **/Certificate/Create** - Create new certificate (Admin/Analyst)
- **/Certificate/Edit/{id}** - Edit certificate (Admin/Analyst)
- **/Dashboard** - Analytics dashboard (Admin/Analyst)

Admin Routes

- **/Admin/UserManagement** - User CRUD operations (Admin only)
- **/Admin/RoleManagement** - Role assignment (Admin only)
- **/Admin/ProductManagement** - Product management (Admin/Analyst)
- **/Admin/Tags** - Tag management (Admin/Analyst)
- **/Admin/CertificateTypes** - Certificate type config (Admin only)

API Endpoints

- **GET /api/v1/certificates** - List certificates with pagination
- **GET /api/v1/certificates/{id}** - Get certificate by ID
- **POST /api/v1/certificates** - Create certificate (requires JWT)
- **PUT /api/v1/certificates/{id}** - Update certificate (requires JWT)
- **DELETE /api/v1/certificates/{id}** - Delete certificate (requires JWT + Admin)
- **GET /api/v1/products** - List products
- **GET /swagger** - OpenAPI documentation (http://localhost:5101/swagger)

Change Log Summary

Date	Change	Impact
2025-10-31	Formal requirements document created	Professional deliverable for business stakeholders
2025-10-30	Design consistency (US-014) completed	External/Internal views now visually identical
2025-10-29	Product Management RBAC (US-012) completed	Secured product operations to Admin/Analyst only
2025-10-20	Dashboard export enhancement (US-006B)	Stat cards now export full certificate details
2025-10-06	Certificate image upload (US-003)	Enabled visual certificate previews in tile views
2025-10-06	Infinite scroll implementation	Improved performance for large datasets
2025-10-06	Autocomplete search with categories	Enhanced user search experience
2025-09-28	Certificate download with uploaded files	Fallback to generated PDF if no uploaded file
2025-09-18	Product ID management (100s of ASINs)	Supports Amazon and multi-retailer products
2025-08-27	14-state workflow implemented	Complete certificate lifecycle management

Contact Information

Project Team: GCC Database Solution Development Team

Organization: Green Circle Certified

Document Owner: Project Manager / Technical Lead

Last Review Date: October 31, 2025

Next Review Date: Post-deployment (30 days after launch)

Document Version: 1.0 - In Testing
